

STEPHANIE KRIEWALL

Data Science | Analytics



206-819-1846



slkriewall@gmail.com



linkedin.com/in/skriewall



Thornton, CO

PROFESSIONAL SUMMARY

Versatile data scientist with extensive programming experience, statistical analysis experience, and a proven track record of adapting to new domains. Master's in Statistics with a strong ability to transform complex data into actionable insights that support decision-making. Skilled in data cleaning, visualization, and communicating analytical findings to technical and non-technical audiences.

SKILLS

Python

R

SQL

Data Analysis

Data Visualization

SAS

Git

Perforce

Microsoft Office suite

Tableau

AWS

C++

JavaScript

EXPERIENCE

SOFTWARE DEVELOPER, Greenwood Village, CO

[Mentice](#), May 2019-August 2025

- Developed and maintained core simulation framework for endovascular and neurovascular procedures (VIST product line), enabling hands-on training with real-time physics and medical device interaction. Used by 700+ customers – including hospitals, universities, simulation centers, and top 20 MedTech companies – to support clinical decision-making and improve patient outcomes.
- Created a 3D augmented reality surgical training tool in Unreal Engine, providing immersive visualization of heart procedures for enhanced clinician engagement and learning.
- Solved a critical Unreal Engine adaptation challenge for medical simulation by implementing the centripetal parameterization of Catmull-Rom splines, preventing cusps and self-intersections in tortuous vascular anatomy.
- Researched and developed an adaptive spline simplification algorithm to reduce control point density and vessel mesh complexity while preserving anatomical fidelity, significantly improving runtime performance in high-complexity cases.
- Automated build workflows using Python, including a custom installer that seamlessly integrates Unreal Engine applications with the VIST simulation environment.
- Contributed to the full development lifecycle as part of a cross-functional Agile team, collaborating closely with fellow engineers and 3D designers.

MATHEMATICAL ANALYST, Wheat Ridge, CO

[Gaming Laboratories International](#), March 2015-May 2019

- Developed C++ game simulations and used probability models to analyze the return to player and odds for slot machine games.
- Performed statistical analysis of simulations and conducted RNG randomness testing to ensure regulatory

compliance.

- Evaluated player strategy and house edge for table and slot games using concepts including probability theory, expected value, Markov decision processes, dynamic programming, and game theory.
- Trained new analysts and conducted detailed technical reviews to ensure accuracy and consistency in game evaluations.

ANALYST, Greenwood Village, CO

[FIMAC Solutions LLC](#), November 2014-March 2015

- Conducted prepayment modeling and other statistical analysis to support risk and performance assessments for financial institutions.
- Developed code for integration into company software products.

EDUCATION

MASTER OF SCIENCE IN STATISTICS, Iowa City, IA

[The University of Iowa](#), May 2014

BACHELOR OF ARTS IN MATHEMATICS, Milwaukee, WI

[Wisconsin Lutheran College](#), May 2010